

Assignment 2: Chapters 6-11.2

Due 5pm, 14 March 2026

Answer each of the following questions along with a brief explanation (1- 3 sentences):

1. An HKU astronomer bought a small apartment in Hong Kong with only one window facing East.
 - a. On the day of summer solstice, from his window he sees a zodiac constellation rising just after midnight. What is the approximately elongation of this constellation?
 - b. Nine months later, around what time of day can he see this constellation through his window?
2. One day during daytime, he sees a “crescent” Moon rising from the East.
 - a. What does this tell you about the angular separation between the Moon and the Sun?
 - b. What is the time, approximately?
3. His colleague told him that recently Mars is in retrograde motion. What time approximately will Mars rise?
4. His grandfather has the same birthday as him. However, he uses the Gregorian calendar only and his grandfather uses the Chinese calendar only. This year, their birthdays happen on the same day. If they both will at least have a few decades more to live, will they be able to celebrate birthday together on the same day again? If yes, why? If no, why not?